Homework 4 for Heat Transfer taught me a lot about thermal resistance networks. I think this is helpful to understand for the future because it creates a visual representation of the thermal resistances and temperatures of the system we are solving, making it easier to understand the process as a whole. Additionally, this homework was essential for my understanding of square fins, how they work, and how to solve for temperatures regarding them because before this assignment the concept was not quite clear. This is important because pin fins will be important in my career as a mechanical engineer.

Heat Transfer HW 4

1) • **known**: $t_g = 3\,mm$, $k_g = 1.5\,W/(m\cdot K)$, $t_a = 0.1\,mm$, $k_a = 1\,W/(m\cdot K)$, $\eta \downarrow$ as $T_{si} \uparrow$

$\eta = a - bT_{si}$, $a = 0.553$, $b = 0.001\,K^{-1}$, $R = 10^{-4}\,m^2\cdot K/W$, $G = 800\,W/m^2$

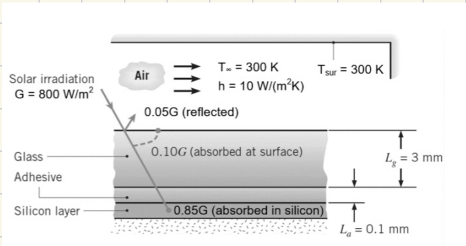
5% rad reflected, 10% rad absorbed by top of glass, 85% rad absorbed by silicon, $T_\infty = 300\,K$, $h = 10\,W/(m^2K)$, $T_{sur} = T_\infty = 300\,K$, $h_{rad} = 5\,W/m^2K$

• **find**: a) thermal resistance network. State key assumptions. Calculate value of each thermal resistance in the network.

b) $T_{g,top}$ & T_{si}

c) η and resulting electricity power flux

• **schematic**:



• **analysis**:

a) - key assumptions:

- silicon layer is isothermal
- constant properties
- steady state

$$h_{tot} = h + h_{rad} = 10 + 5 = 15\ W/(m^2K)$$

$$R_1 = \frac{1}{15}\ m^2K/W$$

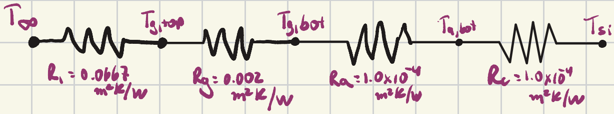
$$R_g = \frac{L_g}{k_g} = \frac{0.003}{1.5} = \boxed{0.002\ m^2K/W}$$

$$R_a = \frac{0.0001}{1} = \boxed{1.0\times10^{-4}\ m^2K/W}$$

$$R_c = \boxed{1.0\times10^{-4}\ m^2K/W}$$

$R_{tot} = \frac{1}{15} + 0.002 + 1.0\times10^{-4} + 1.0\times10^{-4}$

$R_{tot} = 0.0689\ m^2K/W$



b) $\eta = 0.553 - 0.001\, T_{si}$

$q'' = 80 + 680(1-\eta)$

$1-\eta = 0.447 + 0.001\,T \Rightarrow q'' = 80 + 680(0.447 + 0.001T)$

$q'' = 384 + 0.68T = \dfrac{T - 300}{0.0689}$

$384 + 0.68T = \dfrac{T-300}{0.0689} \Rightarrow 26.46 + 0.0469T = T - 300$

$\rightarrow 326.46 = 0.9531\,T \Rightarrow \boxed{T_{si} = 342.5K}$

$q'' = \dfrac{342.5 - 300}{0.0689} = 616.8\ W/m^2$

$\Delta T = q''\, R_{conv} = (616.8)(0.0667) = 41.14K$

$\Delta T = T_{g,top} - T_{g,bot} = 41.14K = T_{g,top} - 300$

$\boxed{T_{g,top} = 341.14K}$

c) $\eta = 0.553 - 0.001(341.14) = 0.553 - 0.341$

$\boxed{\eta = 0.212}$

$P'' = 680\,\eta = (680)(0.212)$

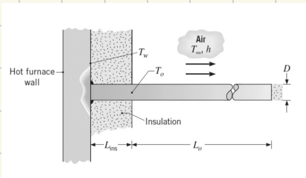
$\boxed{P'' = 144.16\ W/m^2}$

2) • **known**: $D = 10\,mm$, $k = 80\ W/m\cdot K$, $T_w = 200°C$, $L_{ins} = 200mm$, L_o, T_o
$T_{max} = 100°C$, $T_\infty = 20°C$, $h = 20\ W/m^2K$

• **assumption**: insulated, steady state

• **find**: a) expression for T_o
b) T_o, does it meet operating limit?
c) L_o that ensures T_o meets limit

• **schematic**:



• **analysis**:

a)
$$T(x) - T_\infty = (T_w - T_\infty)\frac{\cosh[m(L-x)]}{\cosh(mL)}$$

$$T_o - T_\infty = \frac{T_w - T_\infty}{\cosh(mL)} \qquad m = \sqrt{\frac{\pi D h}{k\frac{\pi D^2}{4}}}$$

$$T_o - T_\infty = \frac{T_w - T_\infty}{\cosh\left(\sqrt{\frac{4h}{kD}}\,L\right)}$$

b)
$$m = \sqrt{\frac{4(20)}{80(0.01)}} = 10\ m^{-1}$$

$$mL = 10(0.2) = 2 \qquad \cosh(2) = 3.76$$

$$T_o - 20 = \frac{200 - 20}{2.76} \Rightarrow T_o = 20 + 47.9$$

$$\boxed{T_o = 68°C}$$

* meets limit *

c) $T_0 - T_\infty = \dfrac{T_w - T_\infty}{\cosh\left(\sqrt{\frac{4h}{kD}}\,L\right)}$

$100 - 20 = \dfrac{180}{\cosh(10L)}$

$80\cosh(10L) = 180$

$\cosh(10L) = \dfrac{180}{80} \Rightarrow \dfrac{\cosh^{-1}(2.25)}{10} = L$

$\hookrightarrow \dfrac{1.45}{10} = L = 0.145\text{m}$ $\boxed{L = 145\text{mm}}$

3) • <u>known</u>: $k = 400$ W/mK, $s = 10$ mm, $h = 100$ W/m²K $\eta_f = 0.6$

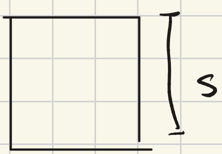
• <u>asumptions</u>: square fin, convection, steady state

• <u>find</u>:

a) fin length L

b) thermal resistance R_f & effectiveness ε_f

• <u>schematic</u>



• <u>analysis</u>:

a) $m = \sqrt{\dfrac{hP}{kA}} = \sqrt{\dfrac{(100)(4\cdot 0.01)}{(400)(0.01)^2}} = 10 \text{ m}^{-1}$

$\eta_f = 0.6 = \dfrac{\tanh(10L)}{10L} \Rightarrow 0.6 = \dfrac{\tanh(x)}{x}$

(through trial and error) $\rightarrow \dfrac{\tanh(1.48)}{1.48} = 0.61$

$x = 1.48 = 10L$ $L_c = 0.148 \Rightarrow L_c = L + \dfrac{s}{4} \Rightarrow 0.148 = L + 0.0025$

$\boxed{L = 0.146 \text{ mm}}$

b) $R_f = \frac{1}{\eta_f h A_f} = \frac{1}{(0.6)(100)(0.04)(0.146)} = \frac{1}{0.3504} = \boxed{2.85\ K/W = R_f}$

$\varepsilon_f = \frac{\eta_f A_f}{A_b} = \frac{(0.6)(0.04)(0.146)}{0.0001} = \boxed{35.04 = \varepsilon_f}$